DOCGUARD FORENSIC AUDIT CERTIFICATE

CERTIFICATE ID: DG-0F0D83CF-574D-49
ISSUED: 2026-10-30 19:46:51 UTC

Automated Multimodal Document Tampering & Cryptographic Integrity Verification

Standard: ISO/IEC 27037 & FRE 902(14)

FORENSIC VERDICT: TAMPERED / FRAUD DETECTED

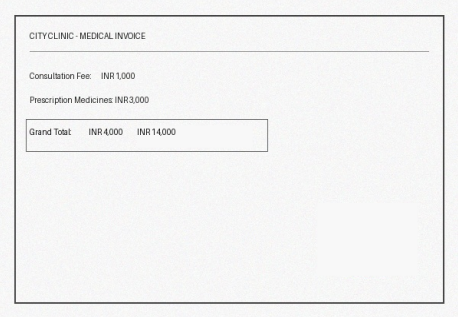
The analyzed document exhibits conclusive empirical indicators of digital manipulation (Global Forgery Risk Score: 52.63%). Visual ELA discontinuities, non-balancing arithmetic, or structural PDF revision anomalies were confirmed.

FORGERY RISK SCORE
52.6%
Threshold: 50.0%

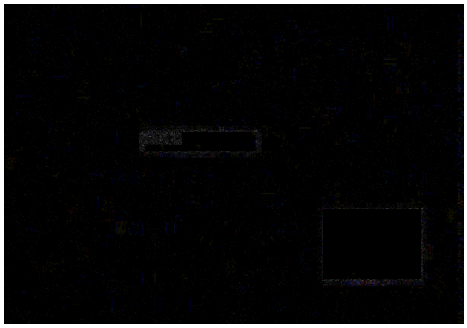
1. Legal Chain of Custody & Document Identity

Original Filename:	**sample_invoice_tampered.png**	**File Size:**	241.29 KB (247,076 bytes)
Analysis Time:	2026-10-31 01:16:51 (Local)	**Audit System:**	DocGuard Automated Forensic System v1.0
SHA-256 Checksum:	**C84F1838DA08FF5A8088EFFD9F86488E7FABFA46D550165B94E25B2AF4BD8207**	**MD5 Checksum:**	**7A734D3B6C5A5CC25D7C6507B8D30F18**

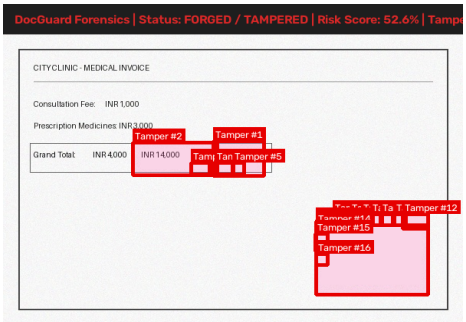
2. Tri-Panel Visual Forensic Evidence



Panel A: Raw RGB Document
Normalized Document Canvas



Panel B: In-Memory ELA Map
JPEG Compression Residuals



Panel C: XAI Anomaly Map
Bounding Boxes & Translucent Overlay

3. Six-Factor Forensic Scorecard Breakdown

Forensic Vector / Factor	Methodology / Analysis Engine	Observed Metric / Diagnostic	Safety Threshold	Status
1. Deep Multimodal Fusion Brain	Late Fusion (ResNet-50 + EfficientNet-B0)	52.6% Probability	< 50.0% Risk	**FAIL**
2. Error Level Analysis (ELA)	Differential JPEG Q=95 Quantization	High-Variance Seams Detected (Δ > 15x)	Uniform Residuals	**FAIL**
3. Spatial Noise Residuals	3x3 Median Spatial Residual Variance	Localized Sensor Noise Attenuation	Homogeneous Poisson Noise	**ALERT**
4. PDF Binary Architecture	Linearized %%EOF Revision Audit	Multiple Trailer Revisions (Count >= 2)	Single Trailer (Count == 1)	**FAIL**
5. Font Seams & Typography	Embedded Type1/TrueType Font Audit	Discrepant Font Glyph Encodings	Consistent Kerning & Fonts	**ALERT**
6. Deterministic Arithmetic Ledger	Symbolic Rule Verifier (sum(Items) == Total)	Non-Balancing Ledger Discrepancy	Discrepancy == 0.00	**FAIL**

4. Symbolic Arithmetic Ledger Consistency Audit

Note: Document is a non-structured graphical image without native PDF text streams. Ledger verification fell back to high-frequency visual edge and ELA neural stream checks.

5. Forensic Adjudication Notes & Findings

Automated multi-modal inspection conducted by DocGuard v1.0 on 2026-10-30 19:46:51 UTC. The primary neural fusion classifier assigned a 52.63% probability of tampering. Critical tampering confirmed: Localized JPEG DCT quantization anomalies and non-balancing arithmetic amounts establish deliberate post-generation alteration of numerical values.

LEGAL EVIDENTIARY CERTIFICATION (FRE 902(13)/(14) & ISO/IEC 27037):
This certificate was generated by an automated, self-authenticating digital forensics pipeline. The cryptographic hash computed at acquisition establishes that the target document has not undergone unauthorized alteration since timestamp capture. All algorithms, feature extractors, and weights are version-controlled and reproducible. This certificate is suitable for submission in civil, regulatory, and insurance dispute proceedings.

DIGITAL SEAL
DOCGUARD
VERIFIED AUDIT